DATA BASES – PROF. D. M. BRAGA – A.Y. 2021/22

CALL 2 – FEBRUARY 10TH 2022 – DURATION: 1h 30m

SQL (24 points)

The schema below describes usage and maintenance of all the public swimming pools in Lombardy. The categories of tickets may be *all-day*, *evening-only*, *student-discount*, *over-65*… with variable prices (possibly even free).

CUSTOMER (CustID, Name, City, BirthDate)
SWIMMINGPOOL (PoolID, Name, Address, City, OpeningDate, Capacity)
TICKET (TickID, CustID, PoolID, EntranceDate, Category, Price)
MAINTENANCE (PoolID, MaintenanceDate, Activity)



1. Briefly but precisely explain what is extracted by the following queries [**5 p**]:

a.
```
select *
from CUSTOMER C
where BirthDate = ALL
   ( select OpeningDate
     from SWIMMINGPOOL natural join TICKET
     where CustID = C.CustID )
```

b.
```
select *
from CUSTOMER C
where not exists ( select *
                   from SWIMMINGPOOL
                   where City = C.City
                     and ( C.CustID, PoolID ) not in ( select CustID, PoolID
                                                       from TICKET )
```

2. Extract in SQL:

- **a**. the Name of the *youngest* customer of the *oldest* pool (i.e., the pool that opened first); [**5 p**]
- **b**. the customers who went to a public pool on more than five of their birthdays. [**5 p**]
- **c**. the *wrong* tickets, if any (i.e., tickets issued for maintenance days or dates before the opening of the pool); [**5 p**]

3. The following **materialized** view tracks, for each City, the number of issued tickets and the revenue for each distinct available category of tickets:

```
create view YAMV( Cit, Cat, TickNo, Revn ) as
    select City, Category, count( * ), sum( Price )
    from TICKET natural join SWIMMINGPOOL
    group by City, Category
```

Assuming that tickets are always inserted (sold) and never deleted (reimbursed) nor updated, formulate a set of triggers that incrementally maintain the materialization in response to *insertions* on table Ticket. Please note that only categories with at least one sold tickets are included in the view (there are no "zero tuples"). [**4 p**]

Conceptual Design (8 points)

4. Design an E-R model for the database described in the following scenario:

A physical rehabilitation center provides diagnostic and physiotherapy services. Patients are visited by a doctor, and when the pathology has been ascertained (diagnosis) the doctor produces a prescription containing the pathology and the description and duration (in days) of the physiotherapy cycle. At this point the patient is assigned a physiotherapist and all the planned time slots (date and time) for the therapy. For all the people involved, we know SNN, name, age and address. Some patients may get to the center with external prescriptions, issued by external doctors (whose name and SSN are recorded). The database also records the actual attendance to the gym room and the therapies performed. Patients may be accommodated in the center (if they live far away from the center or have dense schedules), or may show up at the gym at the prescribed time slots. Patients staying at the center are also characterized by a room and bed number, floor number, and possibly some dietary restrictions.

- *The project is evaluated for completeness, correctness, legibility, minimality and self-explanatory nature.*
- *Remember to specify the **identifiers** of all entities and the **cardinalities** of all associations.*